parties. But in a decentralized system, there are no humans checking identities or banks verifying credentials. You need some other mechanism--a software mechanism. Blockchain addresses this need by enabling a trust layer over the internet, thus facilitating trust intermediation. The three Ds--decentralized infrastructure, distributed ledger technology, and disintermediation protocol--collectively enable trust in a system.

> **NOTE** In a decentralized system, trust intermediation is achieved by the decentralized infrastructure (section 1.3.1), the DLT (section 1.3.2), and the disintermediation protocol (section 1.3.3).

Figure 1.11 shows the evolution of the protocols leading to blockchain-based trust, which has yet to become a standard in the internet context.

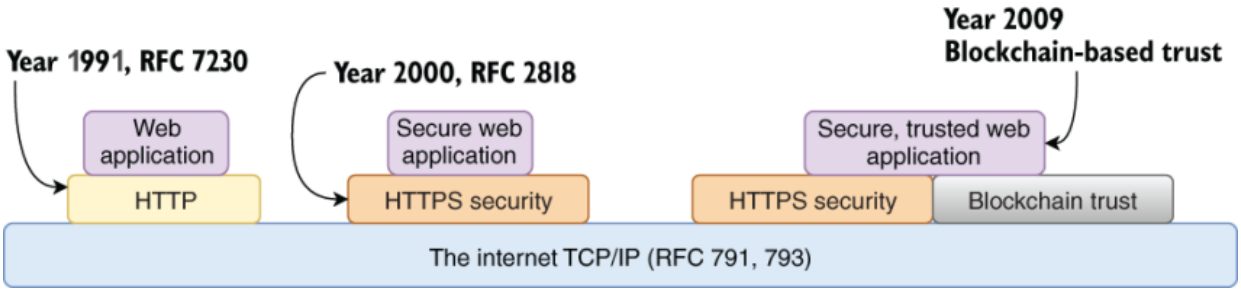


Figure 1.11 Evolution of the internet and the blockchain-based trust layer

The internet was created for sharing research among scientists. It enabled connectivity among computing machines and internetworking. Later, Hypertext Transfer Protocol (HTTP) was introduced as the underlying protocol for the web. It became a standard around 1991 and opened many commercial activities through web applications.